

Permutation rearranges foreground pixels; it never changes how many

One representative packet, binarized at threshold 0.4 (fitted max_value_ = 255); foreground pixel count beneath

Clean



69 / 1500

baseline

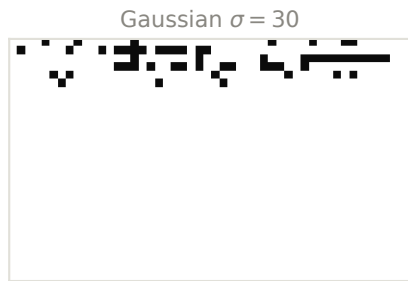
Permuted



69 / 1500

$\Delta = 0$

Noised



56 / 1500

$\Delta = -13$

Illustrative single packet, selected as the jointly most typical of the 200 targets (smallest combined standardised deviation of clean count and noise $|\Delta|$ from their population means).

Population: clean 62.305/1500 · permutation Δ count 0/200, all four families · noise mean $|\Delta|$ 14.02.